Jacob Horne

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Education

University of California, Irvine **Expected Graduation: June 2028**

Bachelor of Science in Computer Science; Campuswide Honors Collegium *GPA: 3.92*

Coursework: Data Structures & Algorithms, Operating Systems, Distributed Systems, System Design, Databases, Machine Learning, Artificial Intelligence, Computer Architecture, Software Engineering, Information Retrieval

Experience

Sandia National Laboratories Aug 2026 – Present

Software Engineer Intern | Data Infrastructure & AI Security *Livermore, CA*

- Developing Databricks workflows for CISA cloud telemetry across heterogeneous security data sources and formats
- Implementing custom parsers and input validation for diverse cloud telemetry formats to improve data reliability
- Creating agentic benchmarks for hidden software triggers; using Docker and Linux to reproduce and debug failures

Capital One Jun 2026 – Aug 2026

Software Engineer Intern | ML Recommender Infrastructure *New York, NY*

- Built Databricks/PySpark sampler for **800M+ daily outcomes**, enabling **28x faster** testing with **18 checks** and Airflow
- Optimized PySpark workloads with partitioning, broadcast joins, caching to minimize data shuffling and recomputation
- Migrated ML routing to an in-process SDK, preventing timeouts and reducing latency **25%** for **60M+ monthly users**
- Wrote unit/E2E tests for production ML routing and offline evaluation, validating with Postman and AWS CloudWatch

SENS Psychology Sep 2025 – Jan 2026

Software Development Intern | Platform and Automation *Arlington, VA*

- Launched Next.js/TypeScript intake and ticketing platform used by **30+ staff** across **1,200+ monthly patient interactions**
- Reduced patient intake time **50%** and manual emails **75%** using Node.js/PostgreSQL automation and Azure notifications

Commit the Change at UCI Oct 2025 – Jun 2026

Full-Stack Developer | Backend and Platform *Irvine, CA*

- Built React/TypeScript scheduling and quota platform with staff access controls supporting **8,000+ patients/year**
- Developed Node.js/Express/PostgreSQL provider directory serving **24+ providers**, saving **12+ staff hours/week**

Research

UCI Digital Learning Lab Dec 2025 – Present

Research Assistant | LLM Systems and Evaluation *Irvine, CA*

- Analyzed token log-probabilities with Python/NumPy to assess LLM confidence in reasoning and SQL tasks
- Co-authored **3 publications** on LLM confidence and reasoning at **ICLR Workshop**, **ACL SRW**, and **GEM** (2026)
- Evaluating BIRD-Interact multi-turn text-to-SQL using execution feedback, SQL repair, and token confidence signals

California Institute for Telecommunications and Information Technology Apr 2026 – Present

Machine Learning Research Assistant | Physics-Informed Machine Learning *Irvine, CA*

- Built a physics-informed PyTorch/GRU prediction pipeline to replace computationally expensive thermal simulations
- Generated thermal predictions **12,000× faster** than traditional simulations, achieving $R^2 = 0.99$ and **MAE = 1.54°C**

Projects & Open Source

PyTorch - Open Source Contributions | *Dynamo* Sep 2026 – Present

- Fixed **3+ bugs** in Dynamo's compiler infrastructure, adding regression tests to improve graph compilation correctness

Agonus - AI Agent Trading Platform | *Blockchain at UCI* Oct 2025 – Apr 2026

- Built FastAPI/Next.js/PostgreSQL services and LangChain agents for autonomous trading, execution, and planning
- Designed asynchronous Celery/Redis task scheduling, distributed locking, and crash recovery for **10+ concurrent agents**
- Engineered PostgreSQL schemas using SQLAlchemy and Pydantic for trade data modeling, validation, and persistence

EthicChat - Ethics Gated Financial Chatbot | *Tata Consultancy Services* Jan 2026 – Jun 2026

- Built React/FastAPI RAG assistant with PDF parsing, OpenAI embeddings, and pgvector for grounded Q&A
- Integrated DeBERTa ethics gating, document classification, and human escalation to control unsafe model responses

Technical Skills

Languages: Python, C++, Java, SQL, TypeScript, JavaScript, C, R, Bash
Backend/Web: FastAPI, Node.js, Express.js, PostgreSQL, Redis, Celery, REST APIs, React.js, Next.js, pgvector
Data/ML: Databricks, PySpark, Airflow, PyTorch, scikit-learn, TensorFlow, NumPy, Pandas, LangChain, Matplotlib
Systems/Cloud: Linux, Docker, Git, CI/CD, Azure, AWS ECS, Fargate, S3, CloudWatch, PowerShell, Postman